

Who Should Be in the Loop?

Expertise-Aware Review for LLM-Assisted Qualitative Coding

Anonymous ACL submission

Abstract

Large language models can assist qualitative coding, but different source-grounded flaws may require different reviewers. We present **WarrantRoute**, an expertise-aware design that routes a theme-and-evidence packet to qualitative-methods review, domain review, both, or neither. RQ1 tests whether flaw-specific human routing increases serious-error recall without increasing false acceptance relative to expertise-agnostic routing. RQ2 defines a controlled-variant evaluation of role-prompted LLM reviewer proxies whose model snapshot and prompts must be frozen before execution; any comparison with human expertise remains contingent on RQ1. Dreddit supports development and an in-domain RQ2 audit; CaChE provides the held-out evaluation for RQ1 and RQ2. The current manuscript reports the protocol and verified corpus preparation only; no review or routing experiment has run. [ADD THE ANALYZED SAMPLE, PRIMARY RQ1 EFFECT, PRE-HUMAN RQ2 RESULT, AND BOUNDED PRACTICAL IMPLICATION FROM THE FROZEN REAL-CORPUS EXPORTS].

Manuscript skeleton: gray fields and blank table cells must be replaced from the frozen analysis export.

1 Introduction

In many forms of qualitative analysis, researchers assign codes, concise descriptive or interpretive labels, to passages that bear on a research question. A codebook organizes the evolving set of codes and, when the analytic approach calls for it, records their definitions, boundaries, and examples. Creating, comparing, revising, and applying these structures across many documents requires substantial interpretive labor. Large language models (LLMs) are therefore being explored as aids for drafting candidate codes (Dai et al., 2023), organizing related codes (Wang et al., 2026), and applying

an established coding scheme across large text collections (Matveyenko et al., 2026). Researchers often review these outputs, edit suggestions, explain disagreements, and guide later analysis. Specific systems operationalize that input differently: Dai et al. (2023) ask a human coder to revise generated codes and supply rationales; CentaurTA collects binary labels and natural-language rationales from human annotators to refine its prompts (Wang et al., 2026); and Muse lets users revise coding instructions and turn positive or negative annotation feedback into examples for later runs (Matveyenko et al., 2026). These studies neither compare reviewer groups on the same source-grounded items nor test whether routing an output to flaw-relevant expertise improves error detection.

This distinction is central to the routing problem studied here: different flaws may require different kinds of expertise to detect. A qualitative methods expert may recognize that a topic label has been presented as an interpretive theme; a domain expert may notice a lost cultural distinction or an omitted clinically important exception; and a researcher without either specialization may still identify an unsupported quotation or an overly broad claim. If these reviewer groups detect different problems, collapsing their judgments into a single average score obscures when each kind of review adds value.

Current evaluations use qualitative expert review and automatic similarity metrics (Yi et al., 2025), aggregate code-space coverage (Chen et al., 2026), human-LLM agreement (Liu et al., 2026), or self-reported confidence (Li et al., 2026). Agreement may diverge from blind expert preference, and self-reported confidence may be poorly calibrated or anchor-sensitive. Audits have also found missing participant evidence, summaries presented as quotations, and failures that depend on local context (Parkington et al., 2025; Bang et al., 2026; Subbiah et al., 2026). Provenance identifies the source of a

passage. It does not show that the passage supports the interpretation or represents the range of views in the corpus. The question is not simply whether review is needed, but *who should be in the loop for each LLM-assisted qualitative coding output?*

We ask:

RQ1. Does flaw-specific routing to human qualitative-methods and domain experts increase serious-error recall without increasing false acceptance, relative to expertise-agnostic routing?

RQ2. Before human review, how accurately do frozen, role-prompted LLM reviewer proxies detect the same controlled flaws; once RQ1 is complete, how do they compare with the corresponding human roles?

Under the protocol, **WarrantRoute** routes each packet to no additional review, a qualitative-methods role, a domain role, or both. RQ1 fills these roles with humans. RQ2 requires role prompts and an LLM reviewer snapshot frozen before execution and applies them to the same packets and rubric. Its first stage evaluates controlled variants against frozen defect labels after all data, privacy, and model-processing approvals are complete; this supports a pre-human flaw-detection result, not a claim that LLMs possess or replace human expertise. Figure 1 summarizes the framework. This paper contributes two study designs: (1) a source-grounded test of whether human reviewer specialization improves flaw detection and (2) a matched evaluation of LLM reviewer proxies frozen before execution, with any human-substitution claim deferred until RQ1 is complete. The present manuscript reports the protocol and verified corpus preparation only; it reports no experimental result. **[REPLACE THIS STATUS SENTENCE WITH THE PRIMARY REAL-CORPUS FINDING ONLY AFTER THE FROZEN STUDY IS COMPLETE].**

2 Related Work

Human-in-the-loop systems let analysts revise codes and rationales (Dai et al., 2023), discuss disagreements (Gao et al., 2024), and convert repeated feedback into persistent instructions (Wang et al., 2026). These systems establish the value of interaction, but they do not test whether qualitative-methods and domain experts detect different source-grounded flaws on the same items.

Evaluator identity can change conclusions: blind expert preference may diverge from agreement metrics (Liu et al., 2026), and source-reading specialists may differ from nonspecialists who see isolated themes (Moskalewicz et al., 2026).

Selective-review methods route items using confidence, ensemble diversity, or rationale disagreement (Zhao and Liu, 2025; Tajik et al., 2026), although confidence can be miscalibrated or anchor-sensitive (Li et al., 2026). Provenance systems link quotations, codes, and themes (Ye et al., 2026; Yi et al., 2026), but a valid link does not ensure that the excerpt warrants the interpretation or preserves relevant counterevidence. WarrantRoute joins these two lines of work: it routes a source-linked packet according to the kind of review it appears to require, then tests the same bounded task with humans and frozen LLM reviewer proxies.

3 Datasets

The protocol assigns complementary roles to two real corpora (Table 1). Dreddit’s official training split supports development and its test split is reserved for an in-domain audit; all 11 Cache2 (CaChe) focus groups are reserved for cross-domain evaluation. “Held out” describes only this study’s development procedure, not model-pretraining exposure.

Dreddit. Dreddit is an author-distributed English Reddit stress corpus containing 3,553 labeled segments from 2,929 posts across ten subreddits and five domains (Turcan and McKeown, 2019). Its official train and test splits determine the study roles. Because the release lacks author identifiers, analyses preserve post clusters and make no participant-level claims.

Cache2 (CaChe). Cache2 is a version-1 CC BY 4.0 deposit of 11 focus groups on COVID-19, schooling, and sexual and reproductive health in western Kenya (Mehta, 2024). The source study reports 54 adolescent girls and young women and 53 community men; sessions used English, Swahili, Dholuo, or mixtures and were translated into English (Awiti et al., 2024). The project retains participant and collective turns separately and does not treat the published themes as evaluation ground truth.

These preparation inventories are not approved experimental samples. Model processing and rater display remain blocked until corpus-specific insti-

Corpus	Context	Language	Verified preparation inventory	Protocol role
Dreaddit (Turcan and McKeown, 2019)	Stress-related Reddit posts	English	2,929 posts; 3,553 released segments; 3,516 content-rule candidates (2,804 train; 712 test); 301,279 words	Train: development; test: in-domain audit
Cache2 (CaChE) (Mehta, 2024)	Western Kenya focus groups	English translations	11 groups; 3,944 retained turns; 3,076 content-rule candidate participant turns (98,735 words)	Held-out cross-domain evaluation

Table 1: Deterministic real-corpus preparation statistics and assigned protocol roles. Dreaddit candidate counts reflect duplicate handling and complete-post exact-date quarantine; CaChE candidate counts apply the three-word participant-turn rule. These are not privacy-cleared samples, final packet denominators, or experimental outcomes.

tutorial, source or platform, provider, privacy, and access requirements are complete; all development choices must be frozen before held-out generation. Every displayed excerpt and context window requires two-person privacy and permissions review. Complete provenance, leakage controls, and access conditions appear in Appendix B.1.

4 Methods

4.1 Evidence packets and controlled variants

The unit of analysis is a **theme and evidence packet**. The packet specification requires each authorized real-data packet to contain a qualitative claim, a short explanation, and exact excerpts only from an access-controlled, pseudonymized analysis copy after the required two-person privacy and permissions review, with project-pseudonymous speaker or document identifiers, local context, and a summary of source coverage. The rating task asks whether the evidence supports the claim and preserves important differences, counterevidence, and source voice. The task is a bounded source-warrant and representation assessment. It excludes a complete reflexive thematic analysis and does not assume that one thematic account is universally correct (Braun and Clarke, 2021a,b).

The sampling specification includes natural LLM outputs and variants vetted by experts. By design, each variant changes one property fixed in the preregistration:

1. **unsupported evidence**: a topically related excerpt does not warrant the claim;
2. **source concentration**: wording implies breadth while evidence comes disproportionately from one participant or document;
3. **counterevidence loss**: a rare, negative, or boundary case is removed;

4. **contextual flattening**: a consequential cultural or domain specific qualification is erased; or

5. **unsupported abstraction**: a local observation is generalized beyond its evidence.

The construction protocol requires an independent panel to check the original packet, the intended defect, and the absence of a second manipulation. Pilot items are excluded from the main analysis, and each rater sees at most one version of a base packet.

4.2 Packet ratings

The common instrument presents the same blinded packet and response form to every reviewer. It asks reviewers to identify the target flaw, choose accept, revise, reject, or escalate, and rate whether the excerpts warrant the claim, preserve consequential differences, and support its stated scope. The primary routing outcomes are serious-error recall and false acceptance; the ordinal packet ratings diagnose why errors were or were not detected. “Cannot judge” remains a separate response. Full anchors, deterministic source checks, and secondary artifact-quality analyses appear in Appendix B.4.

4.3 Human and LLM reviewers

RQ1 is designed to compare researchers without specialist expertise, qualitative-methods experts, and domain experts. The common instrument and assignment design give all groups the same tutorial, blinded packets, and response form with balanced subsets. The assignment protocol requires methods and domain experts to rate independently before adjudication and keeps defensible disagreement visible rather than forcing one answer.

RQ2 requires prospectively frozen generalist-, methods-, and domain-role prompts with a separately identified LLM reviewer snapshot. The ac-

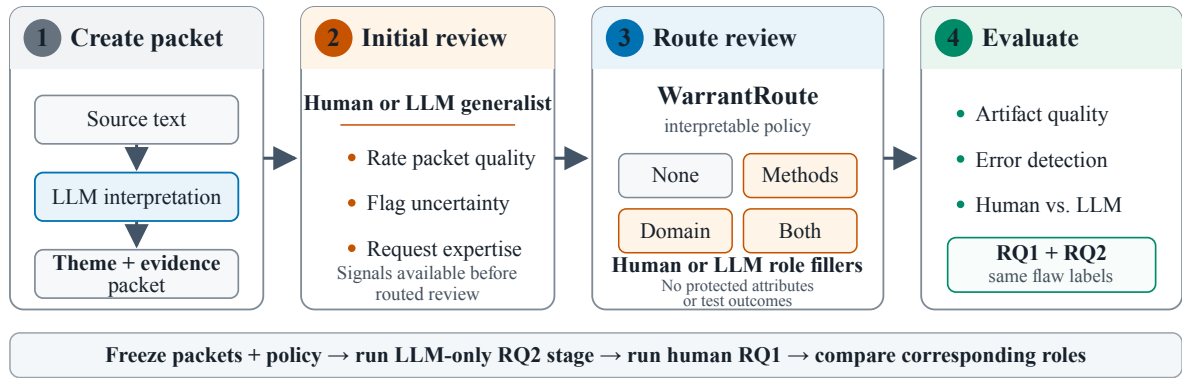


Figure 1: Overview of the WarrantRoute evaluation. An interpretable policy routes each source-grounded packet to no additional review, a qualitative methods role, a domain role, or both. RQ1 uses human reviewers; RQ2 uses role-prompted LLM reviewer proxies on the same bounded task.

tor registry requires distinct records for candidate-generation and reviewer models, and the reviewer payload excludes defect labels, human responses, adjudication, and outcomes. The protocol requires all LLM outputs to be sealed before collection of human ratings.

Before recruitment or model runs, the preregistration must fix recruitment criteria, sample size, LLM repetitions, exclusions, and actor manifests. Full role definitions, assignment, prompts, and failure handling appear in Appendix B.2.

4.4 Routing by Reviewer Role

The policy uses only information available before specialist review: an initial rating, confidence, abstention or requested expertise, and nonsensitive packet-provenance features. It excludes defect labels, protected attributes, specialist scores, and test outcomes. An interpretable regularized model chooses no additional review, methods review, domain review, or both. The protocol requires development-only threshold selection and freezing before held-out evaluation.

The two main comparators are random routing and uncertainty-only routing. Researchers-only review and review by both expert groups are descriptive reference conditions reported in the appendix. Human policy comparisons receive matched specialist-review time as a design control, not as a research question. RQ2 applies the frozen routing task to LLM reviewer proxies and reports their latency, tokens, and cost separately from human time. The protocol requires one fixed-role LLM comparator, selected using development data only and frozen before held-out evaluation.

4.5 Evaluation and Analysis

RQ1: human expertise-aware routing. The primary outcomes are serious-error recall and false acceptance. The primary analysis estimates WarrantRoute’s difference from random and uncertainty-only routing overall and by flaw family, with packet and reviewer clustering. A held-out useful-role label tests whether the policy selected helpful expertise and never enters the router.

RQ2: role-prompted LLM review. The pre-human analysis scores frozen LLM reviewers and routes against locked controlled-variant labels, reporting target-flaw recall, acceptance of defect-bearing packets, abstention, and terminal failures. After RQ1, paired analyses on identical packets and frozen routes compare LLM and human reviewers by role and flaw. Agreement alone does not establish validity or substitution; natural-output and human-equivalence claims require independent human judgments.

The reporting specification requires uncertainty intervals for all primary effects. The appendix contains the full model, multiplicity, missingness, calibration, timing, and sensitivity analyses. Before CaChe processing, actor, prompt, packet, and analysis manifests must be frozen.

5 Results

No review or routing result is yet available. Every placeholder below must be populated from its frozen, authorized real-corpus analysis export. Report the CaChe evaluation before the Dreddit audit, reconcile every denominator, and identify the preregistration, actor snapshots, prompt hashes,

run dates, and analysis freeze. Study accounting and secondary artifact-quality results belong in Appendix D.

5.1 RQ2: Pre-human LLM review

[REPORT TARGET-FLAW RECALL AND ACCEPTANCE OF DEFECT-BEARING PACKETS BY PROMPTED ROLE AND FLAW FAMILY]. [REPORT THE DIFFERENCE BETWEEN WARRANTROUTE AND THE DEVELOPMENT-SELECTED FIXED-ROLE LLM COMPARATOR WITH 95% INTERVALS]. Report abstentions and terminal failures with exact reviewer snapshot and prompt identifiers. This stage does not support a claim about human substitution.

5.2 RQ1: Human expertise-aware routing

[REPORT SERIOUS-ERROR RECALL AND FALSE ACCEPTANCE FOR WARRANTROUTE, RANDOM ROUTING, AND UNCERTAINTY-ONLY ROUTING]. [REPORT THE TWO PRIMARY DIFFERENCES WITH 95% INTERVALS AND THE LARGEST ROLE-BY-FLAW PATTERN]. Paired human-LLM contrasts are reported only after both RQs are complete.

Figure 2 reserves fields for role-by-flaw estimates, and Table 2 reserves fields for the primary corpus-level contrasts.

6 Discussion

[STATE THE RQ1 ROUTING EFFECT, INTERVAL, FALSE ACCEPTANCE, AND MAIN ROLE-BY-FLAW PATTERN]. [STATE THE PRE-HUMAN RQ2 RESULT AND, ONLY AFTER RQ1, THE PAIRED HUMAN-LLM DIFFERENCE]. The central interpretation is whether matching a flaw to a reviewer role improves detection. RQ2 measures prompt-conditioned LLM behavior on the same bounded task; it does not establish that an LLM possesses human expertise. Timing, cost, calibration, and robustness remain secondary.

7 Limitations

The inferential scope is limited to the recruited reviewers, frozen models and prompts, two English or English-translated corpora, and five flaw families. Role prompting cannot reproduce human

training, lived experience, accountability, or interpretive judgment. Controlled variants isolate single defects, whereas natural outputs may contain subtler or interacting errors, and routing may create false reassurance for items that are not escalated. Finally, this task assesses bounded evidence packets, not complete qualitative analysis, which also requires familiarization, memoing, reflexivity, and accountable interpretation.

8 Ethics Statement

Real-text model processing and rater exposure begin only after institutional, source, provider, access, retention, and release requirements are documented; approval of the original collection does not transfer to this study. Restricted copies are minimized, protected text goes only to approved services, and every excerpt shown to raters or released receives a two-person privacy and permissions review. Raters participate voluntarily, and public artifacts contain only aggregate outputs and specifically cleared material. Details appear in Appendix C.

9 Conclusion

WarrantRoute tests one central claim: source-grounded flaws may benefit from different reviewer roles. RQ1 evaluates human expertise-aware routing; RQ2 evaluates frozen LLM reviewer proxies on the same bounded task and permits a human comparison only after RQ1 is complete. The current manuscript establishes the protocol and reports no experimental result. [AFTER THE FROZEN STUDIES: RQ1 EFFECT AND INTERVAL; PRE-HUMAN RQ2 RESULT; HUMAN-LLM DIFFERENCE IF AVAILABLE; BOUNDED PRACTICAL IMPLICATION].

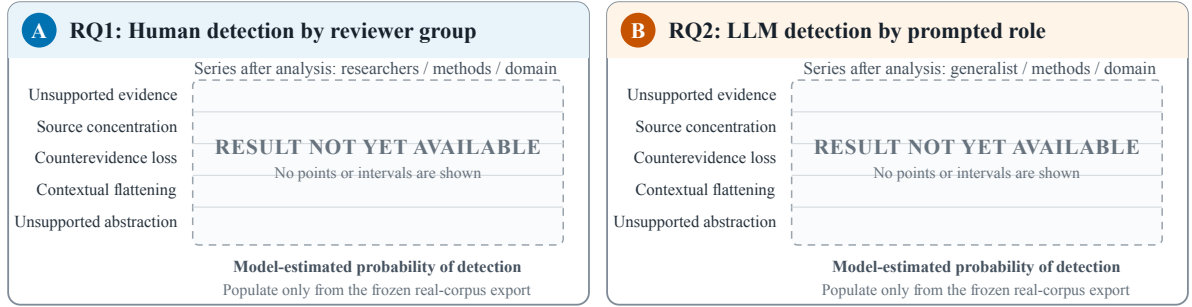


Figure 2: Primary role-by-flaw reporting shell. Panel A is reserved for human detection by reviewer group for RQ1; Panel B is reserved for detection by frozen LLM reviewer prompt for the pre-human stage of RQ2. Points and 95% intervals may be added only from the corresponding frozen real-corpus exports. No experimental results are reported.

Placeholder only—no experimental results are available					
Stage	Corpus	Comparator	<i>N</i>	Recall difference [95% CI]	False acceptance difference [95% CI]
RQ2 pre-human	CaChe	Routed vs. development-selected fixed-role LLM review	[N]	[Δ [CI]]	[Δ [CI]]
RQ2 audit	Dreaddit test	Routed vs. development-selected fixed-role LLM review	[N]	[Δ [CI]]	[Δ [CI]]
RQ1	CaChe	Human routing vs. random routing	[N]	[Δ [CI]]	[Δ [CI]]
RQ1	CaChe	Human routing vs. uncertainty-only routing	[N]	[Δ [CI]]	[Δ [CI]]

Table 2: Primary reporting shell for the two adopted corpora. For RQ2, recall is target-flaw recall and false acceptance is acceptance of a defect-bearing controlled variant. For RQ1, the corresponding outcomes use independently adjudicated serious errors. Populate a row only after its preregistered analysis is complete.

References

- AMI Corpus. n.d. AMI corpus ethics and consent forms. <https://groups.inf.ed.ac.uk/ami/corpus/ethicsandconsent.shtml>. Official corpus documentation; accessed 25 August 2026.
- Enid Awiti, Sophie Young, Garazi Zulaika, Fredrick Odhiambo Otieno, Elizabeth Nyothach, Penelope A. Phillips-Howard, Supriya D. Mehta, and Linda Mason. 2024. “whenever i help her, i am also expecting her vagina in return”: A qualitative analysis to explore men’s and adolescent girls’ perceptions of the impact of the COVID-19 pandemic on the sexual behaviour and health of adolescent girls in rural western kenya. *BMJ Public Health*, 2(2):e001214. Peer-reviewed journal article; open access under CC BY.
- So Hyeon Bang, Soojeong Han, Meghan Reading Turchioe, Melani Ellison, Stacey Dai, Mary Beth Happ, David Russell, and Ruth Masterson Creber. 2026. Developing and evaluating human-led and large language model-assisted hybrid deductive-inductive workflows for qualitative analysis. *Journal of the American Medical Informatics Association*. Peer-reviewed journal article.
- Virginia Braun and Victoria Clarke. 2021a. Can i use TA? should i use TA? should i not use TA? comparing reflexive thematic analysis and other pattern-based qualitative analytic approaches. *Counselling and Psychotherapy Research*, 21:37–47. Peer-reviewed journal article.
- Virginia Braun and Victoria Clarke. 2021b. One size fits all? what counts as quality practice in (reflexive) thematic analysis? *Qualitative Research in Psychology*, 18(3):328–352. Peer-reviewed journal article.
- Jean Carletta, Simone Ashby, Sébastien Bourban, Mike Flynn, Maël Guillemot, Thomas Hain, Jaroslav Kadlec, Vasilis Karaiskos, Wessel Kraaij, Melissa Kronenthal, Guillaume Lathoud, Mike Lincoln, Agnes Lisowska, Iain McCowan, Wilfried Post, Dennis Reidsma, and Pierre Wellner. 2006. The AMI meeting corpus: A pre-announcement. In *Machine Learning for Multimodal Interaction*, volume 3869 of *Lecture Notes in Computer Science*, pages 28–39. Springer.
- Kushal Chawla, Jaysa Ramirez, Rene Clever, Gale Lucas, Jonathan May, and Jonathan Gratch. 2021. CaSiNo: A corpus of campsite negotiation dialogues for automatic negotiation systems. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3167–3185. Association for Computational Linguistics.
- John Chen, Alexandros Nikolaos Lotsos, Sihan Cheng, Lexie Zhao, Yanjia Zhang, Jessica Hullman, Bruce Sherin, Uri Wilensky, and Michael Horn. 2026. A computational method for measuring open codes in qualitative analysis. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 41740–41758. Peer-reviewed archival Findings paper.
- Shih-Chieh Dai, Aiping Xiong, and Lun-Wei Ku. 2023. LLM-in-the-loop: Leveraging large language model for thematic analysis. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9993–10001. Peer-reviewed archival Findings paper.
- Jie Gao, Yuchen Guo, Gionnive Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, and Simon Tangi Perreault. 2024. CollabCoder: A lower-barrier, rigorous workflow for inductive collaborative qualitative analysis with large language models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. Peer-reviewed CHI full paper.
- James Anthony Hale, Sushrita Rakshit, Kushal Chawla, Jeanne M. Brett, and Jonathan Gratch. 2025. KODIS: A multicultural dispute resolution dialogue corpus. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 12771–12785. Association for Computational Linguistics.
- Hongming Li, Huan Kuang, and Anthony F. Botelho. 2026. Can we trust AI’s self-assessment? evaluating and improving LLM confidence calibration in educational dialogue coding. In *Artificial Intelligence in Education*. AIED 2026 conference work; formal proceedings carry 2027 metadata.
- Alex Liu, Lief Esbenshade, Michael Xiao, Victor Tian, Zachary Zhang, Kevin He, and Min Sun. 2026.

490	Agreement is not quality: Blind expert verification	the Association for Computational Linguistics: ACL	548
491	of human and LLM qualitative coding when hu-	2026, pages 15871–15884. Peer-reviewed archival	549
492	man consensus is not ground truth. <i>arXiv preprint</i>	Findings paper.	550
493	<i>arXiv:2607.28890</i> . Preprint; no peer-reviewed		
494	archival version verified as of 24 August 2026.		
495	Joseph Matveyenko, James Liu, John David Parsons,	Runlong Ye, Oliver Huang, Patrick Yung Kang Lee,	551
496	Ryan A. Brown, Alina I. Palimaru, Vipul Gupta, and	Michael Liut, Carolina Nobre, and Ha-Kyung Kong.	552
497	Prateek Puri. 2026. Development and benchmarking	2026. Reflexis: Supporting reflexivity and rigor in	553
498	of a blended human-AI qualitative research assistant .	collaborative qualitative analysis through design for	554
499	In <i>Proceedings of the 64th Annual Meeting of the</i>	deliberation . In <i>Proceedings of the 2026 CHI Con-</i>	555
500	<i>Association for Computational Linguistics: Industry</i>	<i>ference on Human Factors in Computing Systems</i> .	556
501	<i>Track</i> . Peer-reviewed ACL Industry Track paper.	Peer-reviewed CHI full paper.	557
502	Supriya Mehta. 2024. Cache2 focus group discussion	Seungjun Yi, Joakim Nguyen, Terence Lim, Andrew	558
503	transcripts AGYW and community males . University	Well, Joseph Skrovan, Mehak Beri, YongGeon Lee,	559
504	of Illinois Chicago Figshare. Dataset, version 1, CC	Kavita Radhakrishnan, Leqi Liu, Mia Markey, and	560
505	BY 4.0.	Ying Ding. 2025. Position: Thematic analysis of	561
506	Marcin Moskalewicz, Anna Sterna, Karolina Drozd,	unstructured clinical transcripts with large language	562
507	Kacper Dudzic, Marek Pokropski, and Paula Flores.	models . <i>arXiv preprint arXiv:2509.14597</i> . Preprint;	563
508	2026. Computational phenomenology of borderline	presented at GenAI4Health at NeurIPS 2025; no peer-	564
509	personality disorder: A comparative evaluation of	reviewed archival version verified as of 24 August	565
510	LLM-simulated expert personas and human clinical	2026.	566
511	experts . <i>arXiv preprint arXiv:2508.19008</i> . Work-	Seungjun Yi, Joakim Nguyen, Huimin Xu, Terence	567
512	shop presentation reported; archival peer-review sta-	Lim, Joseph Skrovan, Mehak Beri, Hitakshi Modi,	568
513	tus not independently verified by 24 August 2026.	Andrew Well, Carlos M. Mery, Yan Zhang, Mia K.	569
514	Karisa Parkington, Bazen G. Teferra, Marianne Rouleau-	Markey, and Ying Ding. 2026. Automated thematic	570
515	Tang, Argyrios Perivolaris, Alice Rueda, Adam	analysis for clinical qualitative data: Iterative code-	571
516	Dubrowski, Bill Kapralos, Reza Samavi, Andrew	book refinement with full provenance . <i>arXiv preprint</i>	572
517	Greenshaw, Yanbo Zhang, Bo Cao, Yuqi Wu, Sirisha	<i>arXiv:2603.08989</i> . Preprint; manuscript reports sub-	573
518	Rambhatla, Sridhar Krishnan, and Venkat Bhat. 2025.	mission to AMIA 2026; no peer-reviewed archival	574
519	Human vs. LLM-based thematic analysis for digi-	version verified as of 24 August 2026.	575
520	tal mental health research: Proof-of-concept com-	Zhilong Zhao and Yindi Liu. 2025. A confidence-	576
521	parative study . <i>arXiv preprint arXiv:2507.08002</i> .	diversity framework for calibrating AI judgement	577
522	Preprint; no peer-reviewed archival version verified	in accessible qualitative coding tasks . <i>arXiv preprint</i>	578
523	as of 24 August 2026.	<i>arXiv:2508.02029</i> . Preprint; no peer-reviewed	579
524	Melanie Subbiah, Haaris Mian, Nicholas Deas, Ananya	archival version verified as of 24 August 2026.	580
525	Mayukha, Dan P. McAdams, and Kathleen McKeown.		
526	2026. Whose story gets told? positionality	A Appendix Roadmap	581
527	and bias in LLM summaries of life narratives .	The appendix is intentionally skeletal until the pro-	582
528	<i>arXiv preprint arXiv:2604.20131</i> . Preprint; no peer-	protocol is fixed. Complete the subsections below	583
529	reviewed archival version verified as of 24 August	without moving secondary detail back into the eight	584
530	2026.	page main paper.	585
531	Elham Tajik, Conrad Borchers, Bahar Shahrokhian,	B Full Study Protocol	586
532	Sebastian Simon, Ali Keramati, Sonika Pal, and	B.1 Corpus provenance, preprocessing, and	587
533	Sreecharan Sankaranarayanan. 2026. Disagreement	leakage controls	588
534	as data: Reasoning trace analytics in multi-agent sys-	The project documents five candidate corpus lanes.	589
535	tems . In <i>Proceedings of the 16th International Learn-</i>	Dreaddit’s official train partition is provisionally	590
536	<i>ing Analytics and Knowledge Conference</i> . Peer-	assigned to development, its official test partition	591
537	reviewed LAK paper.	to a secondary in-domain audit, and all 11 CaChe	592
538	Elsbeth Turcan and Kathleen McKeown. 2019. Dread-	focus groups to the held-out cross-domain confir-	593
539	dit: A reddit dataset for stress analysis in social me-	matory evaluation. KODIS is prepared only for	594
540	dia . In <i>Proceedings of the Tenth International Work-</i>	a private, unsplit exploratory role. CaSiNo and	595
541	<i>shop on Health Text Mining and Information Analy-</i>	the AMI scenario-meeting subset remain source-	596
542	<i>sis</i> , pages 97–107. Peer-reviewed workshop paper;	inventory dialogue candidates. KODIS, CaSiNo,	597
543	the paper reports no ethics determination or dataset	and AMI have no frozen manuscript experiment	598
544	license.	role, approved model or rater pathway, completed	599
545	Lei Wang, Min Huang, and Eduard Dragut. 2026. Cen-	contextual privacy review, or sampling eligibility.	600
546	taurTA: A self-improving human-agents collabora-		
547	tion framework for thematic analysis . In <i>Findings of</i>		

Dreaddit is a publicly downloadable English Reddit corpus for stress detection. Its released labeled subset contains 3,553 human annotated segments from 2,929 posts across ten subreddits in five domains (Turcan and McKeown, 2019). The archive provides official train and test partitions and post identifiers, but no author identifier. It therefore supports segment to post provenance and document concentration checks, not participant counts or participant level claims. The author archive contains 2,838 train segments from 2,343 posts and 715 test segments from 586 posts. It contains no README or dataset license. We retain text, the stress label, subreddit, and sentence range in the restricted analysis copy. We omit original post identifiers, timestamps, engagement measures, the internally inconsistent confidence field, and unused lexical features. Project specific identifiers preserve post grouping without exposing the released post identifier. The official splits have no post identifier overlap, but three exact texts occur in both partitions under different identifiers and conflicting labels. We retain those rows for audit but make the three training copies and 18 later within train exact duplicates ineligible for sampling, leaving 2,817 development segments and 715 audit segments before privacy quarantine. A checked exact-date rule then quarantines 13 complete development post clusters and three audit clusters, leaving 2,804 development and 712 audit candidate segments, or 3,516 segments and 301,279 whitespace delimited words in total. The remaining text still requires contextual privacy review.

The CaChE deposit contains 11 participant-numbered and selectively redacted focus-group transcripts about how COVID-19 affected schooling, sexual behaviour, and sexual and reproductive health in western Kenya (Mehta, 2024). The source study reports six groups with 54 adolescent girls and young women and five with 53 community men. Sessions used English, Swahili, Dholuo, or a mixture and were translated into English (Awiti et al., 2024). The deposit is version 1, DOI 10.25417/uic.26495884.v1, and CC BY 4.0. A fixed checksum allowlist verifies all 11 source files. Processing retains 3,575 participant turns and 369 collective turns; 3,076 participant turns meet the earlier three-word eligibility rule, but short and collective turns are no longer deleted and remain available as context. Each record has a stable focus-group and turn identifier, a transcript-local speaker identifier, source line spans, source hashes, transla-

tion status, the preceding moderator question, and prior-turn references. Normalization yields 106 observed transcript-local participant keys, one fewer than the 107 participants reported by the source paper; we do not claim complete participant coverage until this is reconciled. Conflicting source dates are recorded but excluded from model and router features.

The received KODIS workbook contains 2,860 unique dyads and 107 columns (Hale et al., 2025). Its 2,085 human-human rows include 2,076 nonempty dialogues with 27,121 parsed messages and 568,333 words; the remaining nine have no transcript. Across all human-human rows, 1,712 are marked Resolution and 373 Impass; among retained transcripts the counts are 1,708 and 368. The median nonempty dialogue has 12 messages (interquartile range 11–14) and 242 words (interquartile range 180–334). The private exploratory builder excludes all 775 human-AI rows and all workbook surveys, demographics, country, preferences, tactics, point totals, raw dyad identifiers, and timestamps. It preserves nine seller-first dialogues and 271 dialogues with at least one consecutive same-role transition rather than imposing artificial alternation. All retained records have the single role exploratory_unsplit; the release has no stable cross-dialogue participant identifier, so it cannot support participant-disjoint splitting or participant-level denominators. This exploratory conversion is not a frozen manuscript experiment or a substitute for contextual privacy review.

The acquired CaSiNo source is the official CC BY 4.0 repository snapshot pinned to commit 2f6ed4a. It contains 1,030 dyadic campsite negotiation dialogues, 11,919 ordinary messages, 2,378 structured negotiation events, and 228,675 whitespace delimited words in the ordinary messages (Chawla et al., 2021). The source paper reports 846 participants, but the public JSON provides only two dialogue local role tokens and no stable cross-dialogue participant token. We therefore reject the supplied 900/30/100 random split for WarrantRoute, do not reconstruct linkage from participant profiles or text, and leave the corpus unassigned unless trustworthy provider linkage becomes available or the entire corpus is preregistered in one unsplit role after the governance gates are satisfied. Any adopted CaSiNo adapter must retain ordinary dialogue text and turn lineage only and exclude structured events, annotations, demographics, personality measures, preferences, free-text

reasons, negotiation outcomes, and raw identifiers.

The acquired AMI manual annotation artifact contains a selectively extracted source inventory for 138 elicited scenario meetings in 35 meeting series. The four person meetings are English product design discussions. The selected files contain 793,764 transcribed words, 69,258 speech segment elements, and mappings for 140 global source speaker identifiers. These form 35 meeting and speaker connected components, 33 with four meetings and two with three meetings (Carletta et al., 2006). No audio, video, signals, slides, questionnaires, demographics, participant file, or nontranscript annotation category was extracted. The official filename labels the manual annotations as version 1.6.2, while the embedded README says release 1.7; we bind the source by its exact URL, filename, size, and checksum rather than asserting that those labels are equivalent. The corpus is not sampling eligible. A adopted AMI transcript adapter must discard timing and raw source identifiers, and all 35 meeting and speaker components must remain indivisible in any adopted split.

For Table 3, a source is one Reddit post, focus group, KODIS human-human dyad, CaSiNo dialogue, or AMI meeting. Dreddit source units are released labeled segments, and its candidate units exclude duplicate texts and complete exact-date post clusters. CaChe retained units include participant and collective turns, while its eligible count applies the earlier three word participant-turn rule. KODIS units are parsed role messages from nonempty human-human transcripts. CaSiNo units are ordinary source messages and exclude structured events. AMI units are upstream speech segment elements. KODIS, CaSiNo, and AMI counts are not sampling eligible or common analysis denominators.

Under this protocol, Dreddit’s official train partition is the sole development corpus for prompts, packet rules, controlled variants, routing features, and thresholds. Once those choices are frozen, its official test partition serves as the in-domain audit, and all 11 CaChe focus groups serve as the held-out cross-domain evaluation. “Held out” refers to the experimental procedure, not model-training provenance. Because the deposit is public and its published themes are known, we do not describe it as unopened or model-unseen or use the published themes as scoring gold.

The power analysis and primary serious-error outcomes determine a common number of base

packets for the Dreddit audit and CaChe evaluation. Human review time is a secondary implementation measure and matched-time comparison control. Sampling preserves post clusters in Dreddit and focus group and transcript local speaker clusters in CaChe. Both corpora use identical packet and context limits. Dreddit labels and published CaChe themes may stratify audits but are not treated as thematic ground truth or router features. KODIS, CaSiNo, and AMI enter no manuscript generation, sampling, or model selection procedure unless separately preregistered after corpus specific validation and governance approval.

Packet generation has not begun and remains blocked until the governance gates are complete. After authorization, packet generation uses [CANDIDATE-GENERATION MODEL SNAPSHOTS] and [GENERATION PROCEDURE]. The protocol requires retention of prompts, decoding settings, context order, run identifiers, source spans, and human revisions and prohibits selection of test outputs after inspection of their quality. Candidate-generation models and LLM reviewers require separate actor identifiers, authorization records, prompts, and run manifests. Only text or transcripts covered by the documented governance arrangement are eligible for processing, and protected material cannot be sent to an external model unless the access terms and our institution permit it. Public release is limited to aggregate outputs, schemas, and text specifically cleared for release.

[CURRENT INSTITUTIONAL DETERMINATION; SOURCE, PLATFORM, CANDIDATE-GENERATION, LLM-REVIEWER, AND HUMAN-RATER AUTHORIZATIONS FOR DREDDIT, CaChe, KODIS, CASINO, AND AMI; KODIS AND CASINO UNSPLIT ROLE DECISIONS; AMI COMPONENT SPLIT; RATER EXCERPT RULES; FINAL PACKET COUNTS; ACTOR REGISTRY; MODEL SNAPSHOTS; PROMPTS; DECODING SETTINGS; RUN COUNTS; AND FROZEN MANIFEST HASHES].

B.2 Role definitions and recruitment

[ELIGIBILITY FOR RESEARCHERS WITHOUT SPECIALIST EXPERTISE, QUALITATIVE METHODS EXPERTS, AND DOMAIN EXPERTS; HANDLING OF DUAL EXPERTISE; RECRUITMENT; COMPENSATION; POSITIONALITY; EXCLUSIONS;

Dataset	Context	Language	Inventory
Dreaddit (Turcan and McKeown, 2019)	Social-media stress posts	English	2,929 posts; 3,553 segments; 3,516 candidate; 301,279 words 11 groups; 3,944 turns;
CaCHe (Mehta, 2024)	Sexual and reproductive health focus groups	Translated English	3,076 eligible; 98,735 words 2,085 human-human dyads; 2,076 dialogues; 27,121 messages;
KODIS (Hale et al., 2025)	Customer-service dispute dialogues	English	568,333 words 1,030 dialogues; 11,919 messages;
CaSiNo (Chawla et al., 2021)	Campsite negotiation dialogues	English	228,675 words 138 meetings; 69,258 segments;
AMI scenario (Carletta et al., 2006)	Product-design meetings	Spoken English	793,764 words

Table 3: Corpus inventories. Dreaddit and CaCHe alone have provisional experimental roles; their counts are preparation inventories, not privacy-cleared samples or experimental outcomes. KODIS, CaSiNo, and AMI are internal candidate inventories and are not sampling eligible.

AND INDEPENDENCE SAFEGUARDS]. For RQ2, [LLM REVIEWER SNAPSHOTS, GENERALIST/METHODS/DOMAIN ROLE PROMPTS, RESPONSE SCHEMA, ONE-PACKET CONTEXT RULE, REPETITIONS, AGGREGATION, RETRIES, TERMINAL-FAILURE HANDLING, PRE-HUMAN OUTPUT-SEAL HASH, AND THE RULE FOR SELECTING ROLE OUTPUTS].

B.3 Complete annotation instrument

The three packet-level rating prompts are: (1) *How fully does the cited evidence support the core interpretive meaning in context?* (evidential credibility); (2) *How well does the packet preserve consequential differences, dissent, counterevidence, less-frequent or boundary cases, and contextual qualifications?* (voice-and-boundary preservation); and (3) *How well do the strength and breadth of the claim, including causal or group-level language, match the supplied evidence?* (scope calibration).

For each question, 1 means not adequate, 2 major problems, 3 partially adequate and requiring material revision, 4 mostly adequate with only minor issues, and 5 fully adequate. The tutorial specification requires construct-specific examples at each anchor. For rating r_{ijm} from rater i on packet j , the analysis specification retains the full ordinal response; the descriptive adequacy rate for measure m is the proportion of evaluable rater–packet judgments scored 4 or 5:

Evidential credibility. A 1 indicates absent, unverifiable, misattributed, or contradictory evidence; a 3 indicates that the core meaning is partly supported but a material link or qualifier lacks evidence; and a 5 indicates that every material part of the core interpretation is directly warranted by sufficient contextual

evidence.

Voice-and-boundary preservation. A 1 indicates that a consequential voice, counterexample, or contextual distinction is erased or reversed; a 3 retains the main pattern but underplays a meaningful viewpoint or boundary; and a 5 retains all consequential variation, dissent, less-frequent cases, and contextual qualifications visible in the packet.

Scope calibration. A 1 indicates that claim breadth, polarity, or causal strength is incompatible with the evidence; a 3 has the broadly appropriate direction but requires a material change in scope or strength; and a 5 fully matches participant-, group-, and corpus-level scope and makes no stronger causal or prevalence claim than the evidence permits.

Scores 2 and 4 represent the corresponding intermediate judgments under the common anchors. Raters select “cannot judge” when the supplied excerpts, local context, or source-coverage summary are insufficient for the specified dimension.

$$A_m = \frac{\sum_{i,j} \mathbb{1}(r_{ijm} \in \{4,5\})}{\sum_{i,j} \mathbb{1}(r_{ijm} \text{ is evaluable})}. \quad (1)$$

“Cannot judge” is neither a midpoint nor a failure and is reported with its own denominator. Confidence has separate anchors from 1 (very uncertain) to 5 (very certain). Exact-string and source-offset checks are recorded as deterministic diagnostics; raters judge whether verified excerpts warrant the interpretation in context.

[CONSENT TEXT, TUTORIAL EXAMPLES, COMPREHENSION CHECKS, ACCEPT/REVISE/REJECT/ESCALATE WORDING, HUMAN CANNOT-JUDGE

RULE, LLM ABSTENTION AND FAILURE RULES, SHARED RESPONSE CONTRACT, RATIONALE PROMPT, AND TIMING DISCLOSURE].

B.4 Artifact-quality definitions and matching

Plural-reference coverage is calculated only for complete, natural-output code and theme inventories, not from sampled packets or controlled twins. The protocol requires multiple analysts to produce code and theme concepts independently and requires the pooled reference space to be frozen before test-output inspection, after merging only concepts adjudicated to be equivalent and retaining the originating analyst and source provenance. The resulting pool is interpreted as a bounded set of defensible concepts available to the study team, not universal thematic ground truth.

For output level $\ell \in \{\text{code, theme}\}$, let P_ℓ be the frozen pooled concepts and G_ℓ the generated concepts. Independent experts blinded to system condition may make many-to-many conceptual matches, denoted by $p \sim g$. Coverage is

$$C_\ell = \frac{|\{p \in P_\ell : \exists g \in G_\ell, p \sim g\}|}{|P_\ell|}. \quad (2)$$

The reporting contract requires separate code and theme coverage with preregistered uncertainty intervals, a list of unmatched reference concepts, retained matching disagreements, and separate evidential review before any unmatched generated concept is counted as an error. It also requires generated code and theme counts, unmatched generated concepts, split and merge patterns, redundancy, and the human-audited unsupported-concept rate. The router cannot access this measure because it relies on the analyst reference space. The three ordinal measures and plural-reference coverage remain separate; no omnibus quality score is permitted.

For controlled variants, the primary manipulation check maps unsupported evidence to evidential credibility; source concentration, counterevidence loss, and contextual flattening to voice-and-boundary preservation; and unsupported abstraction to scope calibration. The analysis also reports cross-construct effects rather than assuming that each defect affects only one measure. Plural-reference coverage is restricted to natural-output inventories and does not serve as a manipulation check for controlled twins.

B.5 Packet construction and assignment

[NATURAL ITEM SAMPLING, CONSTRUCTION OF CONTROLLED VARIANTS, VETTING, MANIPULATION CHECKS, RANDOMIZATION, INCOMPLETE BLOCK ASSIGNMENT, AND BLINDING].

B.6 Power and confirmatory analysis

[PILOT VARIANCE COMPONENTS, SIMULATION DESIGN, POWER TARGET, RQ1 PRIMARY CONTRASTS, RQ2 PAIRED HUMAN-LLM CONTRASTS, MODEL EQUATIONS, MULTIPLICITY PLAN, FALLBACKS, AND EXCLUSIONS].

B.7 Routing implementation

[RQ1 AND RQ2 FEATURE DEFINITIONS, HUMAN- AND LLM-GENERALIST INPUT MAPPING, PREPROCESSING, ROUTING MODEL, HYPERPARAMETERS, FALSE-ACCEPTANCE CEILING, THRESHOLDS, MATCHED-HUMAN-TIME SENSITIVITY, CROSS FITTING, FREEZE DATE, AND TEST ACCESS LOG].

C Detailed Ethics and Data Governance

C.1 Source review and access conditions

Approval of an original collection does not authorize this secondary study, model processing, or disclosure to raters. Real-text processing and rater disclosure remain blocked until the record contains our institution’s determination, its scope and date, and the governing source, platform, and access terms for each corpus.

The received KODIS README records collection dates, field definitions, and a required scholarly citation, but it does not state a data license, secondary-use consent, or authorization for external model processing or rater disclosure. We therefore keep the exploratory output local and do not treat receipt of the workbook as permission for a manuscript experiment (Hale et al., 2025).

The CaChe source study reports written consent and review by Rush University Medical Center (22011311), Maseno University Ethics Review Committee (MSU/DRPI/MUERC/01021/21), and University of Illinois Chicago (2022-0220), with a favorable opinion from Liverpool School of Tropical Medicine (21-087) (Awiti et al., 2024). The CaSiNo paper reports institutional review, informed consent covering collection and later use,

a right to withdraw, and removal of MTurk and HIT identifiers before public release (Chawla et al., 2021). Although that paper calls the released data anonymized, we do not adopt that label: dialogue and rich participant metadata can remain identifying. The AMI site documents European Commission ethics review and signed consent that anticipated possible release to the wider research community; its consent form states that anonymity could not be guaranteed and allowed participants to request removal (AMI Corpus, n.d.). The CC BY 4.0 licenses for CaSiNo and AMI do not grant privacy or personality rights or authorize this project’s model processing, rater disclosure, retention, or release. We did not find an ethics determination, Reddit author consent, privacy procedure, or dataset license in the Dreddit paper or author archive (Turcan and McKeown, 2019); any use therefore remains conditional on a documented institutional and platform basis.

C.2 Project safeguards and rater protections

We do not describe the corpora as anonymous. Dreddit retains searchable verbatim posts. CaChe contains sensitive sexual and reproductive health narratives that may remain identifying despite participant numbers and selective redaction. KODIS combines role-play dialogue with country, survey, preference, tactics, and outcome fields; the private exploratory output retains only dialogue text, minimal row lineage, role, and aggregate diagnostics. CaSiNo raw records combine dialogue with demographics, personality measures, preferences, free text reasons, and outcome surveys; those metadata remain excluded from any authorized working output. AMI source XML contains transcript text, precise timing, and attribute bearing source identifiers; any authorized working output must omit timings and raw identifiers. Restricted analysis copies use project pseudonymous identifiers, minimize fields, mask direct identifier patterns, and preserve protected linkage information separately. Pattern masking does not replace contextual privacy review. Protected text is not sent to a model or service unless its processing, retention, and training terms are approved. [STATE STORAGE LOCATION, ACCESS ROLES, RETENTION PERIOD, DELETION PROCEDURE, APPROVED MODEL PROVIDERS, AND PROVIDER RETENTION OR TRAINING SETTINGS].

No excerpt may be shown to raters or considered for release unless it passes a two-person privacy

and permissions review. Public artifacts are limited to aggregate results, code, prompts, schemas, and only text or ratings specifically cleared for release. Raw hashes of short sensitive excerpts are not published without an assessment of linkage risk. Enrollment requires a study-specific consent form and privacy warning; raters may withdraw under the approved procedure and participate voluntarily outside supervisory, employment, grading, or authorship decisions. The storage policy requires identifiers to remain separate from responses. [STATE RECRUITMENT CHANNELS, ELIGIBILITY, COMPENSATION AND RATE JUSTIFICATION, WITHDRAWAL HANDLING, RATER DEMOGRAPHICS, AND FINAL CONSENT LANGUAGE]. Routing is a research review aid, not a clinical or participant-level decision system; random audits and false-acceptance reporting address, but do not remove, the risk of false reassurance.

D Supplementary Results

The source-grounded artifact profile below is now part of the compiled results shell. Its cells remain blank because corpus inventories cannot substitute for human ordinal ratings, reference-concept matching, or serious-error judgments. No result rows are assigned to KODIS, CaSiNo, or AMI unless a separate preregistration assigns an experimental role.

[ADD ROUTE-TO-USEFUL-ROLE CONFUSION MATRICES, SELECTIVE-RISK VERSUS REVIEW-COVERAGE CURVES, RESULTS BY FAILURE FAMILY, RQ2 LLM-HUMAN CONTRASTS WHEN AVAILABLE, LLM ABSTENTION AND TERMINAL-FAILURE COUNTS, COMPARISONS OF NATURAL AND CONTROLLED ITEMS, CONFIDENCE AND ROUTING-MODEL CALIBRATION PLOTS, RATIONALE ANALYSIS, AND ROBUSTNESS CHECKS].

E Reproducibility and Governance Checklist

- ☐ Confirm institutional, source or platform, model processing, and rater access conditions for Dreddit, CaChe, KODIS, CaSiNo, and AMI; confirm unsplit roles for KODIS and CaSiNo if adopted; validate AMI transcript preprocessing and freeze component isolated splits; then freeze the analytic con-

Corpus / evaluation role	System	Complete inventories	Rated packets	Integrity pass n/N	Evidential adequacy	Voice/boundary adequacy	Scope adequacy	Code coverage	Theme coverage	Cannot judge	Serious error
CaCHe / cross-domain confirmatory	Frozen primary	—	—	—	—	—	—	—	—	—	—
CaCHe / cross-domain confirmatory	Frozen comparator	—	—	—	—	—	—	—	—	—	—
Dreaddit / in-domain audit	Frozen primary	—	—	—	—	—	—	—	—	—	—
Dreaddit / development	Selected primary	—	—	—	—	—	—	—	—	—	—

Table 4: Source-grounded artifact profile. Adequacy is the percentage of evaluable packet ratings scored 4 or 5; “cannot judge” has a separate denominator. Plural-reference coverage is computed separately for complete code and theme inventories. Integrity requires schema and assignment completeness, quotation/source fidelity, valid provenance edges, and cross-source support for cross-source themes. Report 95% intervals clustered by the preregistered sampling unit.

Evaluator group and construct	Krippendorff α	Gwet AC2	Raw agreement	Cannot judge	Median time
Researchers: credibility	—	—	—	—	—
Researchers: voice/boundary	—	—	—	—	—
Researchers: scope calibration	—	—	—	—	—
Methods experts: credibility	—	—	—	—	—
Methods experts: voice/boundary	—	—	—	—	—
Methods experts: scope calibration	—	—	—	—	—
Domain experts: credibility	—	—	—	—	—
Domain experts: voice/boundary	—	—	—	—	—
Domain experts: scope calibration	—	—	—	—	—

Table 5: Reliability shell for human ratings on the three ordinal artifact-quality measures. Report intervals and score prevalence in the final appendix; LLM repetition behavior, confidence, and routing-model calibration are reported separately.

1074	tract, prompts, model snapshots, and mani-	☐ Replace every \fillin, \blankcell, and	1097
1075	fest.	\blankwide; set \predatadraftfalse; ver-	1098
		ify the eight page boundary; and update the	1099
1076	☐ Obtain ethics determination and finalize vol-	abstract last.	1100
1077	untary recruitment and withdrawal procedures		
1078	that avoid coercion.		
1079	☐ Freeze human and LLM role criteria, item		
1080	manifest, actor registry, LLM reviewer snap-		
1081	shots and prompts, the four metric definitions		
1082	and anchors, plural-reference construction and		
1083	matching rules, assignment, exclusions, and		
1084	power analysis.		
1085	☐ Freeze routing features and thresholds, RQ1		
1086	and RQ2 primary outcomes, LLM repeti-		
1087	tions and failure rules, the secondary matched-		
1088	human-time analysis, and analysis code be-		
1089	fore candidate generation or LLM rating on		
1090	CaCHe.		
1091	☐ Release only packets cleared by the two-		
1092	person excerpt review, permitted ratings,		
1093	prompts, code, model and run metadata, non-		
1094	text manifest hashes, and text-derived hashes		
1095	only when source terms permit and linkage-		
1096	risk review clears them, plus a datasheet.		

Placeholder only—no experimental results are available					
Policy	Expert minutes	Serious-error recall	False acceptance	Difference from both experts	Errors per expert hour
Researchers only	0 (design)	[EST. [CI]]	[EST. [CI]]	[Δ [CI]]	Not applicable
Random	[MIN.]	[EST. [CI]]	[EST. [CI]]	[Δ [CI]]	[EST.]
Uncertainty routing	[MIN.]	[EST. [CI]]	[EST. [CI]]	[Δ [CI]]	[EST.]
WarrantRoute	[MIN.]	[EST. [CI]]	[EST. [CI]]	[Δ [CI]]	[EST.]
Both expert groups	[MIN.]	[EST. [CI]]	[EST. [CI]]	Reference by design	[EST.]

Table 6: Secondary matched-human-time reporting shell for the RQ1 CaChE routing comparison. No experimental results are reported. Populate estimates and intervals only from the frozen confirmatory analysis export. Zero specialist minutes for the researchers-only condition and the reference status of both-expert review are protocol design constants, not observed outcomes.

Routing configuration	Dreaddit dev.	Dreaddit audit	CaChE	Recall change	Review minutes
Full features	—	—	—	—	—
No disagreement	—	—	—	—	—
No abstention	—	—	—	—	—
No provenance	—	—	—	—	—
Confidence only	—	—	—	—	—

Table 7: Secondary corpus and routing-feature sensitivity shell. Define the reported metric in the final caption; no experimental results are reported.

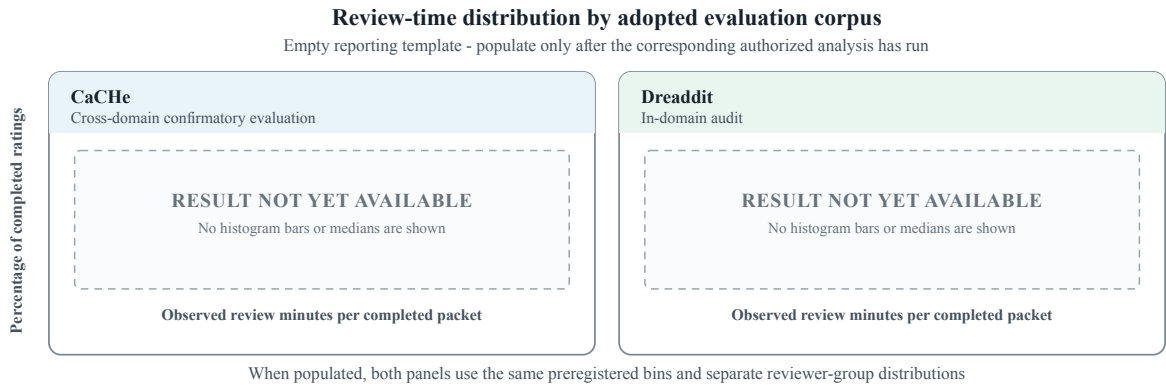


Figure 3: Secondary histogram reporting template for observed human review time per completed packet by reviewer group and adopted corpus. Review time is not a research question, and no experimental results are reported. When populated from the frozen RQ1 export, both panels use the same preregistered bins and show the percentage of completed ratings; median lines are derived from those observations. LLM latency, tokens, and cost are reported separately.